

MINIMAL-SLACK NO-REPEAT SHIFTS: CAYLEY MIXING, A SHORT-PERIOD DESERT, AND EXACT COLOR RETURNS

ANONYMOUS

ABSTRACT. For $q \geq 3$, let N_q be the shift on q colors in which no color may repeat at any of the next $q - 2$ sites. The constraint is one step short of forcing a periodic permutation, and this minimal slack has three exact consequences. We identify the $(q - 1)$ -block presentation with a directed Cayley graph on S_q generated by a $(q - 1)$ -cycle and a q -cycle. Hence N_q is mixing, has entropy $\log 2$, and has a unique measure of maximal entropy. Its short periodic data are sharply nonmonotone:

$$\# \text{Fix}(\sigma^n) = 0 \ (n \leq q - 2), \quad \# \text{Fix}(\sigma^{q-1}) = \# \text{Fix}(\sigma^q) = q!,$$

and, when $q \geq 4$, followed by a second desert through $2q - 3$. Under the maximal measure, the first return of a fixed color is $(q - 2) + \text{Geom}(1/2)$, and the conditional return generating function is $(2 - z)/(2 - z - z^{q-1})$. The least positive period is $q - 1$, so the parameter is recovered from periodic data. Independent state-graph and literal cyclic word enumerations verify the finite controls registered with the note.

1. INTRODUCTION

Local exclusion usually becomes harder as the exclusion window grows: the natural block presentation expands rapidly, and its periodic points become a graph-coloring problem. This note treats the boundary case where an alphabet of q colors is constrained only across the next $q - 2$ positions. Every legal $(q - 1)$ -block then consists of distinct colors, leaving exactly one missing color and exactly two possible continuations.

That one unit of slack produces more structure than the degree count alone suggests. Completing a block by its missing color turns the state space into S_q ; the two continuations become two long position cycles. This precise directed Cayley graph, including its Hamiltonicity, was studied by Ruskey and Williams [4]; it is background here, not a claimed graph-theoretic contribution. The same constraint makes short periodic words equivalent to colorings of powers of a cycle, exposing two consecutive nonzero period counts and a sharply delimited initial zero ledger. Finally, the two continuation labels become fair independent choices under the Parry measure, and the return of one color reduces to a delayed geometric clock.

Shifts of finite type, their Parry measures, and their zeta functions are standard tools [1, 2]. Cycle powers and their colorings are also established objects [3]. After subtracting those owners, the result proved here is the coupled dynamical package for the no-repeat family: its exact initial period ledger, parameter recovery, and rational color-return renewal law. A bounded search through 28 August 2026 did not identify a direct prior treatment of that coupled package; the search is not a priority proof, and we make no absolute novelty claim.

2. THE SHIFT AND ITS PERMUTATION PRESENTATION

Fix $q \geq 3$ and write $\mathcal{A}_q = \{0, 1, \dots, q - 1\}$.

Definition 2.1. The minimal-slack no-repeat shift is

$$N_q = \{x \in \mathcal{A}_q^{\mathbb{Z}} : x_i \neq x_{i+j} \text{ for } 1 \leq j \leq q - 2 \text{ and every } i \in \mathbb{Z}\},$$

with the left shift σ .

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Every legal word of length $q - 1$ has distinct letters. Such a word $w = (a_1, \dots, a_{q-1})$ has a unique missing color a_q . We identify the state w with the bijection $p_w : [q] \rightarrow \mathcal{A}_q$ whose one-line notation is

$$\hat{w} = (a_1, a_2, \dots, a_{q-1}, a_q) \in S_q,$$

where we regard S_q as the set of such bijections.

Proposition 2.2 (Cayley presentation). *In the $(q - 1)$ -block presentation, each state has two successors. Under the completion $w \mapsto \hat{w}$, they are obtained by the position cycles*

$$\alpha = (1 \ 2 \ \dots \ q - 1), \quad \beta = (1 \ 2 \ \dots \ q).$$

Consequently the presentation is a directed Cayley graph of S_q . It is strongly connected and aperiodic.

Proof. After deleting a_1 , the new symbol may be either a_1 or the previously missing color a_q ; every other choice repeats one of the preceding $q - 2$ symbols. Completion of the two new blocks gives, respectively,

$$(a_2, \dots, a_{q-1}, a_1, a_q), \quad (a_2, \dots, a_{q-1}, a_q, a_1),$$

which are exactly $p_w \circ \alpha$ and $p_w \circ \beta$. Thus we use the right-action convention throughout.

The element $\alpha^{-1}\beta = (q - 1 \ q)$ is a transposition. Conjugating it by powers of α , which fixes q and cycles the other positions, gives all star transpositions $(j \ q)$, so α and β generate S_q . This group-generation statement also gives directed reachability: $\alpha^{-1} = \alpha^{q-2}$ and $\beta^{-1} = \beta^{q-1}$ are positive words in the allowed generators. Hence the positive semigroup generated by the two labels is all of S_q , and the directed graph is strongly connected. Repeating the α edge gives a closed walk of length $q - 1$, while repeating the β edge gives one of length q . Their lengths are coprime, so the graph period is one. This is the right-action version, canonically isomorphic after the usual left/right convention change, of the universal-cycle Cayley graph in [4]; we use only the elementary connectivity statement proved above. $\square$

Corollary 2.3 (mixing and maximal entropy). *The shift N_q is mixing and*

$$h_{\text{top}}(N_q) = \log 2.$$

It has a unique measure of maximal entropy μ_q . In its block presentation, μ_q is uniform on the $q!$ states and assigns probability $1/2$ to each outgoing edge.

Proof. The presentation is primitive by [proposition 2.2](#). It is both two-out-regular and two-in-regular, so the all-ones vector is a left and right Perron vector with eigenvalue two. The claims follow from the Parry measure construction [2]. $\square$

3. THE SHORT-PERIOD DESERT

Put $F_q(n) = \# \text{Fix}(\sigma^n|_{N_q})$. An n -periodic point is a coloring of the cyclic positions $\mathbb{Z}/n\mathbb{Z}$ such that positions at positive directed gap $j \in \{1, \dots, q - 2\}$ receive different colors. Equivalently, two distinct positions are adjacent when their shorter cyclic distance is at most $q - 2$; if some forbidden positive gap is 0 modulo n , the graph has a loop and no coloring is allowed. Thus this is the usual cycle-power coloring interpretation, with the small- n loop convention made explicit.

Theorem 3.1 (two islands and a desert). *For every $q \geq 3$,*

$$\begin{aligned} F_q(n) &= 0 & (1 \leq n \leq q - 2), \\ F_q(q - 1) &= F_q(q) = q!, \\ F_q(n) &= 0 & (q + 1 \leq n \leq 2q - 3). \end{aligned}$$

Thus

$$\min\{n \geq 1 : F_q(n) > 0\} = q - 1,$$

and the family parameter is recovered as one plus this conjugacy-invariant minimum.

Proof. If $n \leq q - 2$, the constraint with gap $j = n$ compares each cyclic position with itself, so no periodic point exists. For $n = q - 1$, every pair of positions is constrained and all $q - 1$ colors in the word must be distinct; there are $q!$ ordered choices. For $n = q$, every pair is again constrained: if one directed cyclic distance is $q - 1$, the reverse distance is one. Hence all q positions use distinct colors, in $q!$ ways.

Finally, suppose $q + 1 \leq n \leq 2q - 3$; for $q = 3$ this interval is empty. The shorter cyclic distance between any two positions is at most $\lfloor n/2 \rfloor \leq q - 2$. The cycle power is therefore complete. It would require $n > q$ distinct colors, which is impossible. The recovery statement is immediate. $\square$

The theorem concerns the initial periodic ledger, not all periods. Beyond $2q - 3$ the cycle power is no longer complete and additional periodic points appear. Exact controls below deliberately test only what the theorem claims.

4. EXACT COLOR RETURNS UNDER THE PARRY MEASURE

Fix a color c and condition on $x_0 = c$. Let

$$\tau_c = \min\{n \geq 1 : x_n = c\}$$

be its first positive return time. We use the convention that $G \sim \text{Geom}(1/2)$ has support $\{1, 2, \dots\}$ and $\mathbb{P}(G = k) = 2^{-k}$.

Theorem 4.1 (delayed geometric return). *Under $\mu_q(\cdot \mid x_0 = c)$,*

$$\tau_c \stackrel{d}{=} (q - 2) + \text{Geom}(1/2).$$

Successive return gaps to c are independent with this law. If

$$u_n = \mu_q(x_n = c \mid x_0 = c),$$

then

$$(1) \quad \sum_{n \geq 0} u_n z^n = \frac{2 - z}{2 - z - z^{q-1}}.$$

In particular, for the color indicator $Y_i = \mathbf{1}_{\{x_i = c\}}$,

$$\text{Cov}_{\mu_q}(Y_0, Y_n) = \frac{1}{q} \left(u_n - \frac{1}{q} \right),$$

and the entire correlation sequence is determined by the denominator in (1).

Proof. Under the uniform stationary distribution on states, each outgoing label is chosen with conditional probability $1/2$, independently of the current state. Consequently the sequence of Cayley-edge labels is an i.i.d. fair Bernoulli sequence (and the time-zero state is uniform). The exclusion rule forbids a return during the first $q - 2$ steps. At time $q - 1$, the color c is the oldest color in the current legal block. One of the two equiprobable continuations appends it; the other appends the missing color and makes c the new missing color. Conditional on not returning, at every subsequent step one continuation is c and the other is the oldest block color. The independent fair edge choices from [corollary 2.3](#) therefore give the stated delayed geometric law.

At a return, the return time is a stopping time for the edge-label filtration. The unused i.i.d. labels after that stopping time are independent of its past, and the same state-independent argument restarts. Thus visits to c form a renewal process with independent, identically distributed gaps. Its first-return generating function is

$$f(z) = \sum_{k \geq 1} 2^{-k} z^{q-2+k} = \frac{z^{q-1}}{2 - z}.$$

The renewal identity $u(z) = 1 + f(z)u(z)$ gives (1). Uniformity of the stationary state distribution gives $\mu_q(x_0 = c) = 1/q$, from which the covariance formula follows. $\square$

Remark 4.2. The pole at $z = 1$ in (1) records the limiting value $1/q$. After subtracting it, the other roots of $2 - z - z^{q-1}$ give an exact finite linear recurrence for the correlation decay. No spectral diagonalization of the full $q!$ -state matrix is needed.

5. EXACT CONTROLS AND SCOPE

The registered script constructs all $q!$ block states for $3 \leq q \leq 6$, checks the right-action orientation, the transposition identity, positive-word inverses, both in- and out-degree two, and forward and reverse connectivity. It computes adjacency traces through $2q - 3$ by sparse dynamic programming; a second path enumerates literal cyclic words for $3 \leq q \leq 5$. Finally, exact rational evolution from every eligible marked state is compared with the delayed-geometric first-return law and its two-gap renewal factorization. These are finite falsification controls; the proofs above carry the general statements.

This manuscript is an internal theorem record. Public posting, submission, and priority language remain on hold pending a broader specialist search.

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